

Shi Jie Samuel Tan

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INTRODUCTION

I am a PhD student at the Joint Center for Quantum Information and Computer Science (QuICS) at the University of Maryland, College Park. I am broadly interested in quantum error correction (QEC) and fault tolerance. I utilize concepts and tools in quantum information theory, homological algebra, and graph theory to understand and advance the construction of quantum error correcting codes and fault-tolerant protocols.

EDUCATION

University of Maryland, College Park

Ph.D. in Computer Science; Advisors: Daniel Gottesman and Michael J. Gullans

College Park, MD

2023–2027 (Expected)

Haverford College

B.S. in Computer Science and Physics

Magna Cum Laude; High Honors in Computer Science and Physics

Haverford, PA

2019–2023

WORK

University of Maryland, College Park

Research Assistant; Supervisors: Daniel Gottesman and Michael J. Gullans

College Park, MD

2023–Current

Google Quantum AI

Student Researcher; Mentors: Noah Shutty and Michael Newman

Playa Vista, CA

Fall 2026

IBM Quantum

QEC Theory Research Intern; Mentors: Guanyu Zhu and Theodore J. Yoder

Yorktown Heights, NY

Summer 2026

University of Maryland, College Park

Grader and Teaching Assistant

College Park, MD

2025–2026

QuEra Computing Inc.

QEC Research Intern; Mentors: Hengyun (Harry) Zhou and Arpit Dua

Boston, MA

Summer & Fall 2025

Los Alamos National Laboratory

Quantum Algorithms Research Fellow; Mentors: Yiğit Subaşı and Andrew Sornborger

Los Alamos, NM

Summer 2023

California Institute of Technology

QEC SURF Fellow; Mentors: John Preskill and Christopher A. Pattison

Pasadena, CA

Summer 2022

University of Maryland, College Park

REU-CAAR Summer Researcher; Mentors: Matthew Coudron and Gorjan Alagic

College Park, MD

Summer 2021

PUBLICATIONS & PREPRINTS

♦ denotes equal contribution. ♠ denotes alphabetical order.

- [1] ♠ Matthew Coudron, Michael J. Gullans, Jon Nelson, Joel Rajakumar, and **S.J.S.T.**, “Polynomial-time additive-error estimation of output probabilities for shallow quantum circuits”, [arXiv preprint arXiv:2610.02146](#) (2026).
- [2] Guanyu Zhu, **S.J.S.T.**, Ryohei Kobayashi, and Po-Shen Hsin, “Sipser-Spielman meets Dijkgraaf-Witten: non-Abelian qLDPC codes via twisted sheaf gauge theory and almost-constant-overhead magic state fountain”, [arXiv preprint arXiv:2609.31541](#) (2026).
- [3] Jin Ming Koh, Shayan Majidy, Aranya Chakraborty, Anqi Gong, **S.J.S.T.**, and Norman Y Yao, “Achieving the limits of automorphism gates”, [arXiv preprint arXiv:2609.19250](#) (2026).
- [4] **S.J.S.T.**♦, Ian Gill♦, Eric Huang♦, Pengyu Liu, Chen Zhao, Hossein Dehghani, Aleksander Kubica, Hengyun Zhou, and Arpit Dua, “Generalized matching decoders for 2D topological translationally-invariant codes”, [arXiv preprint arXiv:2603.05402](#) (2026).
- [5] Pengyu Liu, **S.J.S.T.**, Eric Huang, Umut A Acar, Hengyun Zhou, and Chen Zhao, “Achieving Optimal-Distance Atom-Loss Correction via Pauli Envelope”, [arXiv preprint arXiv:2603.04156](#) (2026).
- [6] **S.J.S.T.**♦, Yifan Hong♦, Ting-Chun Lin♦, Michael J Gullans, and Min-Hsiu Hsieh, “Single-Shot Universality in Quantum LDPC Codes via Code-Switching”, [arXiv preprint arXiv:2510.08552](#) (2025), [Accepted as a short plenary talk at QIP 2026].
- [7] ♠ Noah Berthusen, Michael J. Gullans, Yifan Hong, Maryam Mudassar, and **S.J.S.T.**, “Automorphism gadgets in homological product codes”, [arXiv preprint arXiv:2508.04794](#) (2025).
- [8] Noah Berthusen, **S.J.S.T.**, Eric Huang, and Daniel Gottesman, “Adaptive Syndrome Extraction”, [PRX Quantum 6](#), 030307 (2025), [Accepted as a talk at TQC 2025].
- [9] **S.J.S.T.** and Lev Stambler, “Effective Distance of Higher Dimensional HGP’s and Weight-Reduced Quantum LDPC Codes”, [Quantum 9](#), 1897 (2025).
- [10] Dhruvil Patel♦, **S.J.S.T.**♦, Yiğit Subaşı, and Andrew T. Sornborger, “Optimal Coherent Quantum Phase Estimation Via Tapering”, [PRX Quantum 7](#), 020302 (2026).
- [11] **S.J.S.T.**, Christopher A. Pattison, Matt McEwen, and John Preskill, “Resilience of the surface code to error bursts”, [Phys. Rev. A 113](#), 022450 (2026).
- [12] **S.J.S.T.**, Huyen Trang Dang, Sarah Keim, Maja Bucan, and Sara Mathieson, “Identity-by-descent (IBD) segment outlier detection in endogamous populations using pedigree cohorts”, [bioRxiv](#), 2024–08 (2024).
- [13] ♠ Yongtao Deng and **S.J.S.T.**, “Random Walks on the Generalized Symmetric Group: Cutoff for the One-sided Transposition Shuffle”, [arXiv preprint arXiv:2211.10462](#) (2022).
- [14] Suchetan Dontha♦, **S.J.S.T.**♦, Stephen Smith, Sangheon Choi, and Matthew Coudron, “Approximating Output Probabilities of Shallow Quantum Circuits Which Are Geometrically-Local in Any Fixed Dimension”, in [Proceedings of the 17th Conference on the Theory of Quantum Computation, Communication and Cryptography \(TQC 2022\)](#) (2022), [Accepted as a talk at TQC 2022].

- [15] Kelly Finke, Michael Kourakos, Gabriela Brown, Huyen Trang Dang, **S.J.S.T.**, Yuval B. Simons, Shweta Ramdas, Alejandro A. Schäffer, Rachel L. Kember, Maja Bućan, and Sara Mathieson, “Ancestral haplotype reconstruction in endogamous populations using identity-by-descent”, *PLOS Computational Biology*, edited by Degui Zhi, 10.1371/journal.pcbi.1008638 (2021).

TALKS

Invited talks are indicated by ★.

- | | |
|--|--------------------|
| ★ National University of Singapore (NUS) Centre for Quantum Technologies (CQT)
<i>Generalized matching decoders for 2D topological translationally-invariant codes</i> | September 17, 2026 |
| ★ IBM QITEC Summer Workshop
<i>Single-Shot Universality in Quantum LDPC Codes via Code-Switching</i> | June 3, 2026 |
| ★ USYD-PsiQuantum Seminar
<i>Single-Shot Universality in Quantum LDPC Codes via Code-Switching</i> | April 16, 2026 |
| • APS Global Physics Summit
<i>Single-Shot Universality in Quantum LDPC Codes via Code-Switching</i> | March 17, 2026 |
| ★ Harvard University (Lukin Group)
<i>Single-Shot Universality in Quantum LDPC Codes via Code-Switching</i> | March 6, 2026 |
| • 29th Annual Quantum Information Processing Conference (Short Plenary)
<i>Single-Shot Universality in Quantum LDPC Codes via Code-Switching</i> | January 26, 2026 |
| ★ Freie Universität Berlin (Eisert Group)
<i>Single-Shot Universality in Quantum LDPC Codes via Code-Switching</i> | November 18, 2025 |
| • Joint Quantum Institute (JQI) Friday Quantum Seminar (College Park, MD)
<i>Adaptive Syndrome Extraction</i> | October 3, 2025 |
| • 20th Conference on the Theory of Quantum Computation, Communication and Cryptography
<i>Adaptive Syndrome Extraction</i> | September 18, 2025 |
| ★ National University of Singapore (NUS) Centre for Quantum Technologies (CQT)
<i>Automorphism gadgets in homological product codes</i> | May 22, 2025 |
| ★ University of Maryland Communication, Control and Signal Processing (CCSP) Seminar
<i>Effective Distance of Higher-Dimensional HGP's and Weight-Reduced Quantum LDPC Codes</i> | October 17, 2024 |
| • Caltech SFP Summer Seminar Day
<i>Proving the existence of an accuracy threshold for the Union-Find decoder</i> | August 18, 2022 |
| • University of Maryland Research Experience for Undergraduates Combinatorics and Algorithms for Real Problems (REU-CAAR) Research Presentation
<i>Simulation of Low-Depth Quantum Circuits</i> | August 13, 2021 |

POSTERS

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| • 8th International Conference on Quantum Error Correction
<i>Single-Shot Universality in Quantum LDPC Codes via Code-Switching</i> | June 11, 2026 |
| • 8th International Conference on Quantum Error Correction
<i>Generalized matching decoders for 2D topological translationally-invariant codes</i> | June 11, 2026 |
| • 8th International Conference on Quantum Error Correction
<i>Achieving Optimal-Distance Atom-Loss Correction via Pauli Envelope</i> | June 9, 2026 |
| • Monte Verità Quantum Codes
<i>Single-Shot Universality in Quantum LDPC Codes via Code-Switching</i> | April 21, 2026 |

- 29th Annual Conference on Quantum Information Processing
Automorphism gadgets in homological product codes January 26, 2026
- 7th International Conference on Quantum Error Correction
Automorphism gadgets in homological product codes August 12, 2025
- 7th International Conference on Quantum Error Correction
Adaptive Syndrome Extraction August 11, 2025
- 28th Annual Conference on Quantum Information Processing
Effective Distance of Higher-Dimensional HGP's and Weight-Reduced Quantum LDPC Codes February 26, 2025
- 19th Conference on the Theory of Quantum Computation, Communication and Cryptography
Tapered Quantum Phase Estimation September 11, 2024
- 6th International Conference on Quantum Error Correction
Resilience of the surface code to error bursts November 1, 2023
- 13th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics
Comparison of cohort-based identical-by-descent (IBD) segment finding methods for endogamous populations August 7, 2022
- 25th Annual Conference on Quantum Information Processing
Approximating Output Probabilities of Shallow Quantum Circuits which are Geometrically-local in any Fixed Dimension March 10, 2022
- 7th–8th International Conference on Algorithms for Computational Biology
Improvements to ancestral haplotype reconstruction in pedigrees November 9, 2021
- Research in Computational Molecular Biology – 25th Annual International Conference
Improvements to ancestral haplotype reconstruction in pedigrees August 29, 2021
- Haverford College Marian E. Koshland Integrated Natural Sciences Center (KINSC)
Undergraduate Science Research Symposium
Validating Ancestral Haplotype Reconstruction In Endogamous Populations using Identical-by-Descent September 10, 2020

FELLOWSHIPS AND AWARDS

- National University of Singapore Development Grant 2024–2027
- MathQuantum Graduate Fellowship (University of Maryland) 2023
- QuICS Lanczos Graduate Fellowship (University of Maryland) 2023–2025
- Louis B. Green Prize in Physics and Astronomy (Haverford College) 2023
- Phi Beta Kappa 2023
- Los Alamos National Laboratory Quantum Computing Summer School Fellowship 2023
- Singloh Hsu Scholarship (Haverford College) 2023
- Computing Research Association (CRA) Outstanding Undergraduate Researcher Award Finalist 2023
- Caltech Summer Undergraduate Research Fellowship 2022
- Marian E. Koshland Integrated Natural Sciences Center Summer Scholarship (Haverford College) 2022
- An Zhu-Google University of Maryland REU-CAAR Fellowship 2021
- Brian Kovaric Fellowship (Haverford College) 2020
- Prime Minister's Book Prize (Ministry of Education, Singapore) 2016

OTHER ATTENDANCE

- Los Alamos Quantum Computing Summer School June–August 2023

TEACHING (AS TA OR GRADER)

- Computer Science 858G (University of Maryland, College Park) Spring 2026
Quantum Error Correction and Fault Tolerance
- Computer Science 657 (University of Maryland, College Park) Fall 2025
Introduction to Quantum Information Processing
- Physics H302 (Haverford College) Spring 2023
Advanced Quantum Mechanics
- Computer Science B340 (Bryn Mawr College) Fall 2022
Analysis of Algorithms
- Physics H214 (Haverford College) Spring 2022
Introductory Quantum Mechanics
- Computer Science H345 (Haverford College) Spring 2022
Theory of Computation
- Mathematics Question Center Fall 2021–Spring 2022
Calculus, Linear Algebra, and Abstract Algebra
- Physics H213 (Haverford College) Fall 2021
Waves and Optics
- Mathematics H121 (Haverford College) Spring 2021
Multivariable Calculus
- Computer Science H311 (Haverford College) Fall 2020
Computer Security: Attacks and Defenses

REVIEWING

I have been invited to review articles and abstracts for the following conferences and journals.

- Annual Quantum Information Processing Conference (QIP)
- Conference on Theory of Quantum Computation, Communication and Cryptography (TQC)
- Physical Review X Quantum

COMPUTER SKILLS

Ordered from most to least familiar.

- **Languages:** Python, Julia, SageMath
- **QC toolkits:** Stim

REFERENCES

- **Dr. Daniel Gottesman**, Brin Family Endowed Professor in Theoretical Computer Science (Professor)
Joint Center for Quantum Information and Computer Science, University of Maryland, College Park
dgottesm@umd.edu
- **Dr. Michael J. Gullans**, Adjunct Assistant Professor and Director of Quantum Error Correction
Joint Center for Quantum Information and Computer Science, University of Maryland, College Park
QuEra Computing Inc.
mgullans@umd.edu
- **Dr. Hengyun (Harry) Zhou**, Thomas D. and Virginia W. Cabot Professor (Assistant Professor)
Department of Electrical Engineering and Computer Science, Massachusetts Institute of Technology
hyzhou@mit.edu
- **Dr. Guanyu Zhu**, Senior Research Scientist
Quantum Information Theory and Error Correction Team, IBM T.J. Watson Research Center
guanyu.zhu@ibm.com